CUSTOMER CHURN & RETENTION STRATEGY

WEEK 8 CAPSTONE PROJECT

TECHNICAL REPORT

Dataset: Customer Churn | Records Analyzed: 500

Overall Churn Rate: nan%

END-TO-END BUSINESS ANALYSIS • DATA PREPARATION • EDA • STATISTICAL ANALYSIS • PREDICTIVE MODELING

1. EXECUTIVE OVERVIEW

This project presents an end-to-end business analysis of customer churn. The objective is to identify customer characteristics associated with churn, quantify the strength of observed relationships, build a baseline predictive model, and translate the evidence into practical retention actions. The final analytical dataset contains 500 customer records. Overall churn is nan%. The analysis focuses on tenure, monthly charges, contract type, payment method, paperless billing, senior-citizen status and churn.

2. BUSINESS PROBLEM AND OBJECTIVES

Customer churn is a business problem because losing an existing customer can create pressure on retention, service planning and future acquisition requirements. The business question for this capstone is: which observable customer characteristics are associated with churn, and how can those signals support a practical retention workflow?

The project objectives are to establish a reproducible data-cleaning process, explore the customer population, compare churn and retained groups, test statistical relationships, build a baseline classification model, communicate findings clearly, and define an implementation plan that can be measured over time.

3. DATASET DESCRIPTION

The supplied customer churn dataset contains 500 raw records and 9 fields. CustomerID is treated as an identifier rather than a predictive business feature. Tenure represents customer duration, MonthlyCharges and TotalCharges represent billing measures, while Contract, PaymentMethod and PaperlessBilling describe service and billing characteristics. Churn is the target outcome.

The dataset is appropriate for an educational capstone because it supports exploratory analysis, group comparison, categorical association testing and classification. However, it does not contain revenue, profit, acquisition cost, campaign spend or intervention-history fields. Therefore, financial return on retention actions cannot be estimated from the supplied data alone.

4. DATA CLEANING AND PREPARATION

The raw data was inspected for data types, missing values, duplicated customer identifiers and inconsistent numeric representation. Numeric variables were converted into appropriate numeric formats. CustomerID was normalized and duplicate identifiers were removed. Rows with incomplete analytical information were excluded from the final modeling dataset. After preparation, the analytical dataset contains 500 complete customer records.

The cleaning stage is separated from analysis so that the workflow can be reproduced. The repository includes the original raw dataset, cleaned dataset and dedicated data-cleaning notebook. This makes the transformation process auditable rather than hiding preprocessing inside the final analysis.

5. EXPLORATORY DATA ANALYSIS

The overall churn rate is nan%. The most visible difference occurs in customer tenure. Retained customers have an average tenure of nan months, while churned customers average nan months. This indicates that churn is concentrated more heavily among shorter-tenure customers in this sample.

Monthly charges also differ between groups. Retained customers average $nan per month, compared with $nan among churned customers. Contract type provides another important segmentation dimension. The observed churn rates are {'Month-to-month': nan, 'One year': nan, 'Two year': nan}.

6. STATISTICAL ANALYSIS

Statistical testing is used to distinguish observed group differences from patterns that could plausibly arise from sampling variation. Welch's t-test is appropriate for comparing numeric variables across two groups without assuming equal population variances. Chi-square tests are used for categorical associations.

The analysis tests tenure versus churn, monthly charges versus churn, contract versus churn, payment method versus churn, and paperless billing versus churn. P-values are interpreted as evidence about the null hypothesis, not as measures of business impact. A statistically significant association does not by itself establish causation.

7. PREDICTIVE MODELING

A baseline Logistic Regression classifier is used to estimate the probability of churn from selected customer characteristics. Numeric features are standardized and categorical features are one-hot encoded. A stratified train-test split is used so that the class distribution is represented in both partitions.

The model is evaluated using accuracy, precision, recall, F1-score and ROC-AUC. ROC-AUC is particularly useful because it evaluates ranking performance across classification thresholds. The model should be considered a baseline prioritization tool. It should be validated on newer operational data before being used in a production retention workflow.

8. BUSINESS INSIGHTS

Four business insights emerge from the analysis. First, early-tenure customers represent an important retention segment because the average tenure of churned customers is substantially lower than that of retained customers. Second, contract type is strongly associated with churn in the sample, making contract status useful for segmentation. Third, monthly charges differ between churn groups, suggesting that pricing or service-value context deserves further investigation. Fourth, a predictive model can help prioritize customers for human review, but model predictions should not be treated as causal explanations.

9. IMPLEMENTATION PLAN

The recommended implementation is staged. Phase 1 validates the analytical pipeline and model on newer data. Phase 2 creates a weekly risk-prioritization report. Phase 3 strengthens onboarding and engagement for early-tenure customers. Phase 4 tests contract-focused retention interventions. Phase 5 measures outcomes and retrains or recalibrates the analytical workflow.

Success should be measured using actual business outcomes such as churn reduction, retention uplift, intervention response and intervention cost. Controlled experiments should be used where practical so that the business can distinguish intervention effects from ordinary customer variation.

10. TESTING AND VALIDATION

Validation evidence includes checks for row counts, duplicate customer identifiers, missing analytical values, expected categorical values and model evaluation metrics. The notebooks provide executable evidence of the analytical workflow. The separation of raw data, cleaned data, notebooks and reports also supports reproducibility.

Testing should be repeated whenever new data is introduced. A model that performs well on one historical dataset can degrade when customer behavior, pricing, products or acquisition channels change.

11. LIMITATIONS

The analysis has several limitations. The dataset contains 500 customer records, which is sufficient for the educational assignment but smaller than a typical production customer base. The dataset does not include revenue, profit, customer lifetime value, acquisition cost, marketing spend or intervention history. Consequently, financial ROI and causal intervention effects cannot be calculated. The baseline model also requires validation on future data before operational deployment.

12. CONCLUSION

This capstone demonstrates a complete business-analysis workflow from raw customer data to actionable retention planning. The strongest analytical signals in the supplied sample are associated with tenure and contract type, while monthly charges provide an additional segmentation dimension. The project combines descriptive analysis, statistical testing and predictive modeling while clearly separating association from causation.

The next step for a real business would be to connect these analytical signals with intervention history and financial outcomes, then run controlled retention experiments. This would turn the educational analysis into a measurable decision-support system.